THE ART OF CYBERPATRIOT



Goal of this presentation

- Explain classroom guidelines/expectations
- Help you decide which system you should go for
- Outline what is required from you
- Help you decide the best approach to cyber for you
- Define key aspects to achieve maximum performance

Some ground rules

- No games during class
  - We will have breaks!
- Be nice to each other
- Use your inside voices if talking
- Respect your instructors!
  - We spend a lot of time for y’all



Rules con't

- If you are a disturbance to others on multiple occasions, we will ask you to leave the class
- Please respect your fellow students

Who should be here?

- Basic
  - Beginners to the Windows OS and its security features
- Most importantly, people who are WILLING TO LEARN
  - If you’re not willing to learn, please reconsider

What's the point of this class?

What we will be emphasizing

- Security features of Windows
- Research skills
- Better understanding of the system
  - Focus on WHY and HOW, instead of just WHAT is happening
- Concept of networks

Research skills



- Lab guides
  - We'll give you instructions to lead you towards a goal, but we won't tell you the exact steps
  - You have to figure those steps out independently

Better understanding of the system

- Our goal is not focused on grabbing points
- Our goal is focused on learning about the system
  - Those of you who understand the systems the best/spend your time figuring out the intricacies will be the most successful
- If you understand the system, the points will follow

How to get the most out of this course?

We all want to go to nationals don't we?

Just some things we need to make clear

- We instructors are responsible for PROVIDING the info, YOU are responsible for LEARNING it
- This program requires commitment
  - If you aren't willing to invest your time, you won't benefit

General recommendations



- Take notes!!
  - You are free to use additional devices (ex: chromebook) if that helps
    - Ex: could use your chromebook to view the lab instructions/take notes
    - Highly recommended :D
  - Use the power of google docs
    - Also highly recommended :D

General recommendations

- **ASK QUESTIONS!!**
  - We’re here to help you learn
  - But we can’t read your minds
  - If you ever need clarification, don’t be afraid to raise your hand. We won’t judge :)



General recommendations

- Do some independent studying/work
  - Those who succeed in cypat are those who are willing to put in the time!


how win ???
success
time

Google Classroom/Discord



- We will put up the materials for you to reference
  - Slides
  - Lab guides
  - Images



- Discord may be used for extra announcements or information
  - cyber.jmal.xyz



2. Your Destiny

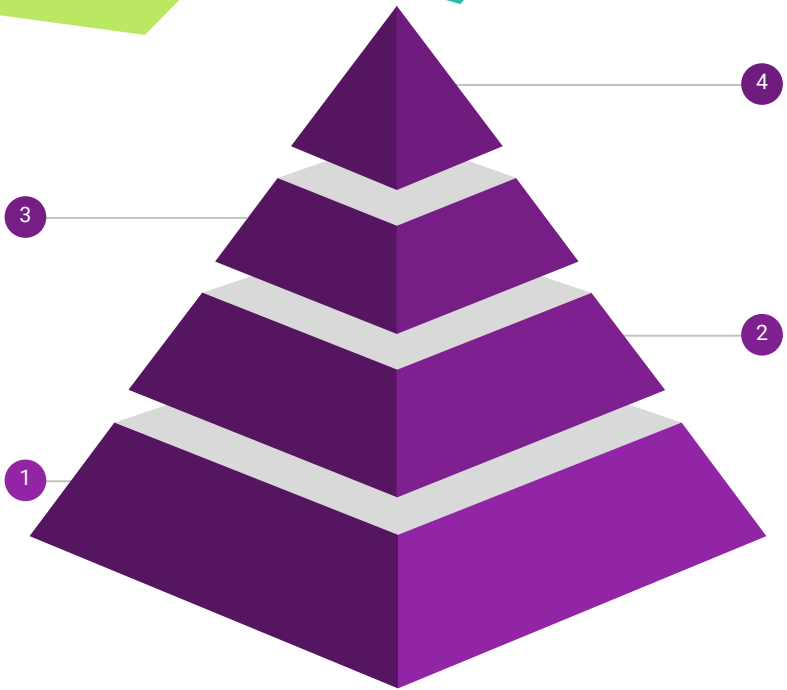
What is the best way to go for **YOU**?

Breadth Approach

Year 1	Year 2	Year 3	Year 4
Attend the classes of any system	Attend the classes of another system	Attend the classes of the last system	Knowing each component of each system, focus on real-life security and combine your knowledge of each system

This approach is not recommended if you intend to compete on a higher level. This is more suited if you wish to focus your time on something else, like Business and Entrepreneurship. In this case, these classes will be more like a supportive elective class that will broaden your general knowledge.

Depth Approach



Senior Year

Take it a bit easier
Improve scripts
Continue research
Teach others to build up a strong legacy

Sophomore Year

Improve your checklist/plan so it is usable for the next years
Work on efficiency and time - find shortcuts
Learn the basics of another system

Junior Year

Research on various topics
Start developing scripts
Teach others to solidify your foundations
Learn the basics of the third system

Freshman Year

Build up fundamentals
Make a detailed checklist/plan

Warning: this approach will plateau the learning curve unless you work hard

Research is 70% of the work in the later years

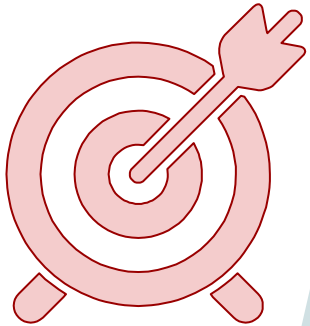
Quick Note

You are not limited to the approaches outlined in the last two slides. However, these are the most common paths taken in our cyber defense program.

3. Intuition and Skill



The part everyone hates - the hard truth



Intuition and Skill



There will be many who have an advantage over you because of their past experiences in the hacking world. If you have never touched a computer, you are at a disadvantage.



Critical analysis and quick response is necessary. You will need to be able to figure something out, even if it hasn't been taught yet.



CyberPatriot also draws in problems that are unsolvable without some CTF experience. CTFs essentially test your ability to apply your knowledge of computers, files, protocols, hacking, etc. to find some message or participate in a hack-and-defend scenario.

Intuition and Skill

If the last slide made you uncomfortable, don’t worry!

You’re here for a reason - to learn!

We will teach you all the fundamentals to help you get started on your journey to Nationals.





4.
Passion

Nothing can be done without some passion

Passion

If you find that you don’t have passion when learning cyber security, you won’t be able to succeed. **Success in CyberPatriot is heavily based on your willingness to research and practice on your own time.**

CyberPatriot is your own investment - “you do for you.”

If you are not here to win, this isn’t the right place for you.



5.
Organization

Being organized is being in control



Organization

Organization is not about perfection.

It’s about efficiency, reducing stress & clutter, saving time and effort, and improving your performance in cyber security.

Meaning GPEDIT

Signal

Computer Configuration > Administrative Templates

Very important !

User Configuration > Administrative Templates

Warning (see side note)

May require change if remote

U

ntested/unsure U Third category Fourth category Fifth category Sixth category Setting Set to Exception Notes Signal Control Panel Allow Online Tips Disabled Personalization Do not display the lock screen Disabled

Force a specific default lock screen and logon image

Enabled (C: \Windows\Web\Scre en\img100.jpg, check box) Force a specific Start background Enabled (1) Prevent changing lock screen and logon image

Enabled

Prevent changing start menu background

Enabled

Prevent enabling lock screen camera

Enabled

Prevent enabling lock screen slideshow

Enabled

Regional and Language Options

Allow input personalization Disabled

Block clean-up of unused language packs

Disabled

Handwriting personalization

Turn off automatic learning Enabled

User Accounts

Enabled

Network

✓	Service ⚙	Windows's Description	Recommended State
	ActiveX Installer (AxInstSV)	Provides UAC validation for the installation of ActiveX controls from the Internet.	Manual
	Adaptive Brightness	Monitors ambient light sensors to detect changes in ambient light and adjust the display brightness.	Manual
	AllJoyn Router Service	Routes AllJoyn messages for the local AllJoyn clients. If this service is stopped the AllJoyn clients that do not have their own bundled routers will be unable to run.	Manual
	App Readiness	Gets apps ready for use the first time a user signs in to this PC and when adding new apps.	Manual
	Application Experience	Processes application compatibility cache requests for applications as they are launched	Manual
	Application Host Helper Service	Handles administrative tasks for Internet Information Services (IIS), Microsoft's web server. This process can be safely disabled if you do not use IIS. It may also be safe to disable if you do not need to control access to dynamic application pools.	Optional Feature in Control Panel
	Application Identity	Determines and verifies the identity of an application.	Manual
	Application Information	Facilitates the running of interactive applications with additional administrative privileges.	Manual
	Application Layer Gateway Service	Provides support for 3rd party protocol plug-ins for Internet Connection Sharing	Disabled
	Application Management	Processes installation, removal, and enumeration requests for software deployed through Group Policy.	Manual (Disabled?)
	AppX Deployment Service	Provides infrastructure support for deploying Store applications. This service is started on demand and if disabled Store applications will not be deployed to the system, and may not function properly.	Manual
	ASP.NET State Service	ASP State Service provides support for out-of-process session states for ASP. ASP has a concept of session state - a listing of values associated with the client session is accessible from ASP pages through the Session property. There are three options provided to store session data: In process, SQL database, and out-of-process. The ASP State Service stores session data out-of-process. The service communicates with ASP using sockets.	Optional Feature in Control Panel
	Background Intelligent Transfer Service	Transfer files using idle network bandwidth, maintain file transfers through network disconnections and computer restarts.	Manual
	Background Tasks Infrastructure Service	Coordinates execution of background work for WinRT applications.	Automatic
	Base Filtering Engine	Manages firewall and IPsec policies and user mode filtering.	Automatic
	Bitlocker Drive Encryption Service	Provides secure startup for theOS, as well as full volume encryption.	Automatic
	Block Level Backup Engine Service	The WBENGINE service is used by Windows Backup to perform backup and recovery operations.	Manual
	Bluetooth Handsfree Service	Enables wireless Bluetooth headsets to run on this computer. If this service is stopped or disabled, then Bluetooth headsets will not function properly with this machine.	Manual
	Bluetooth Support Service	Discovery and association of remote Bluetooth devices.	Disabled
	BranchCache	Caches network content from peers on the local subnet.	Disabled
		Copies user certificates and root certificates from smart cards into the current user's certificate store, detects when a smart card is inserted	

Which would you rather look at?

Organization Methods

- Notes
- Checklists
- Research Documents
- To-Do’s and Links



http://

6.

Team

Two is better than one. Six is better than one.

Teamwork

One of the most important factors towards success is your team.

If your team is not coordinated or motivated, nothing good will come out of it.





7.

Organization

Being organized is being in control



Organization

Organization is not about perfection.

It’s about efficiency, reducing stress & clutter, saving time and effort, and improving your performance in cyber security.

Organization Methods

- Notes
- Checklists
- Research Documents
- To-Do’s and Links



Checklists and notes should only be references; try not to rely on them too much

8.

Notice

important dates etc

ALL CYBERPATRIOT ROUNDS ARE EARLIER

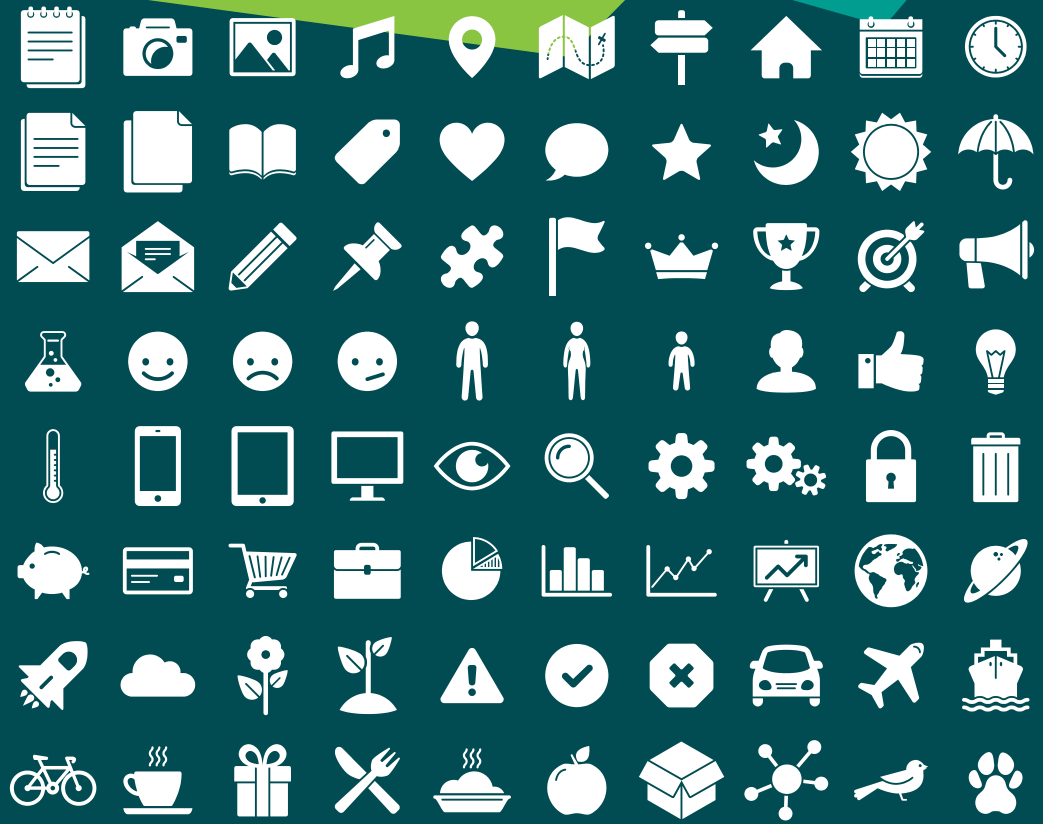
LAST YEAR		THIS YEAR	
11/2-11/4	Round 1	10/25-10/27	Round 1
12/7-12/9	Round 2	11/15-11/17	Round 2
1/11-1/13	Round 3	12/6-12/8	Round 3
2/1-2/3	Round 4	1/24-1/25	Round 4
4/6-4/10	Nationals	3/19-3/23	Nationals

CyberPatriot Promotional Video

https://youtu.be/o159up56-XM

Thanks!

Any questions?



SlidesCarnival icons are editable shapes.

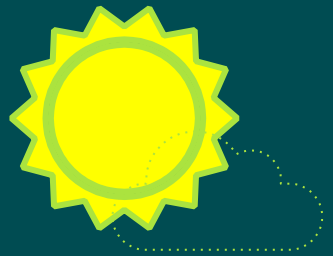
This means that you can:

- Resize them without losing quality.
- Change fill color and opacity.
- Change line color, width and style.

Isn't that nice? :)

Examples:





Now you can use any emoji as an icon!
And of course it resizes without losing quality and you can change the color.

How? Follow Google instructions
https://twitter.com/googledocs/status/730087240156643328



and many more...